

Inventra – Intelligent Inventory Management System

What Is Inventra – Intelligent Inventory Management System ?

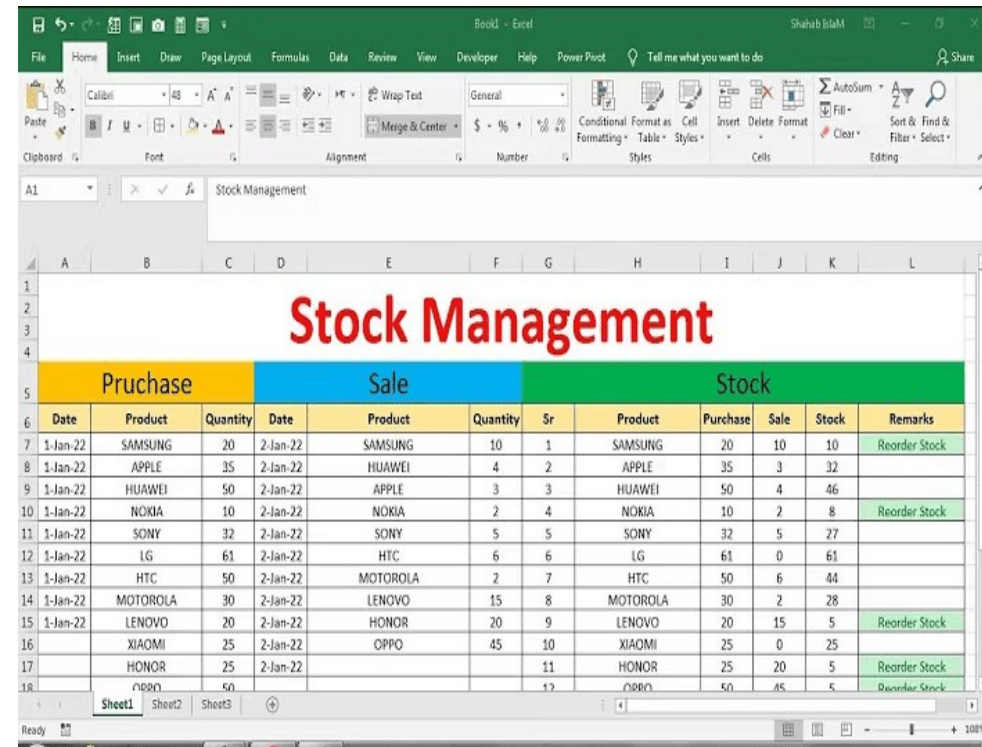
Inventra = Intelligent + Inventory

- A smart system for managing business inventory
- Automates stock tracking and purchase management
- Ensures fast, accurate, error-free inventory operations
- Helps in planning, forecasting & decision-making



Existing System

- Manual registers / Excel sheets used for stock tracking
- No real-time stock updates
- High chance of errors and mismatches
- No alerts for low stock or expiry
- Hard to track fast/slow-moving items
- Time-consuming and unreliable

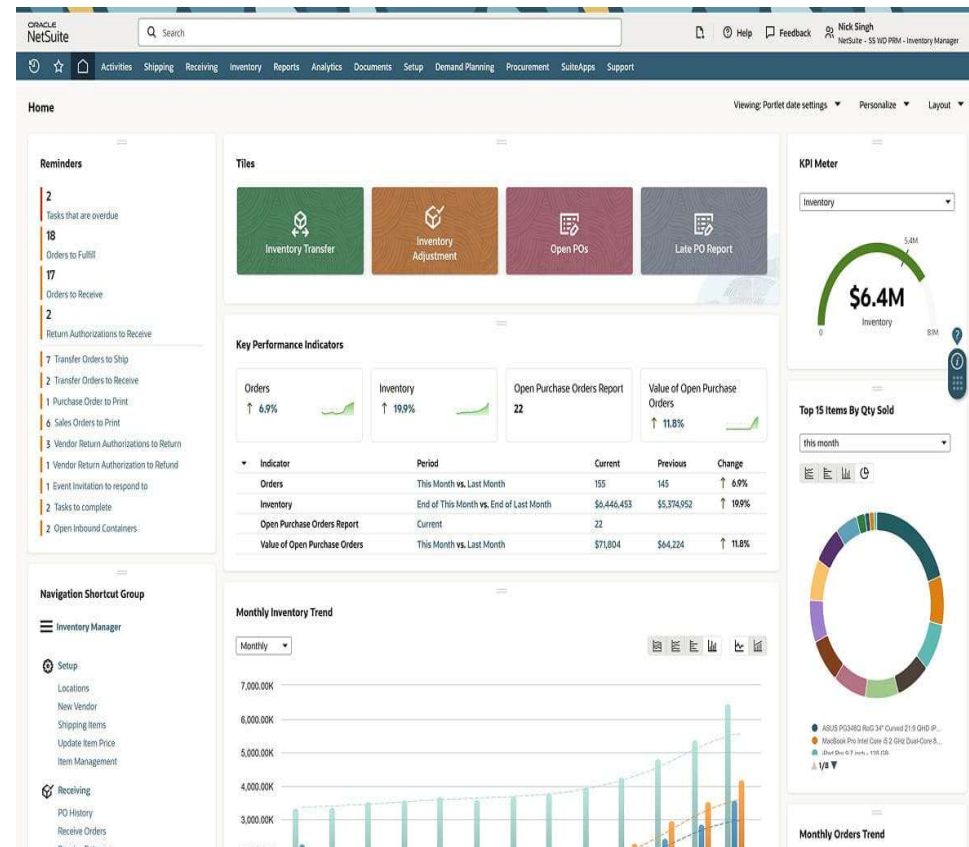


The screenshot shows an Excel spreadsheet titled "Stock Management" with a table containing stock data. The table has columns for Date, Product, Quantity, and Stock levels. The data is organized into three main sections: Purchase, Sale, and Stock. The Purchase section lists products like SAMSUNG, APPLE, HUAWEI, NOKIA, SONY, LG, HTC, and MOTOROLA. The Sale section lists products like SAMSUNG, HUAWEI, APPLE, NOKIA, SONY, LG, HTC, and MOTOROLA. The Stock section lists products like SAMSUNG, APPLE, HUAWEI, NOKIA, SONY, LG, HTC, and MOTOROLA. The table also includes a Remarks column with entries like "Reorder Stock".

Stock Management											
Purchase			Sale			Stock					
Date	Product	Quantity	Date	Product	Quantity	Sr	Product	Purchase	Sale	Stock	Remarks
1-Jan-22	SAMSUNG	20	2-Jan-22	SAMSUNG	10	1	SAMSUNG	20	10	10	Reorder Stock
1-Jan-22	APPLE	35	2-Jan-22	HUAWEI	4	2	APPLE	35	3	32	
1-Jan-22	HUAWEI	50	2-Jan-22	APPLE	3	3	HUAWEI	50	4	46	
1-Jan-22	NOKIA	10	2-Jan-22	NOKIA	2	4	NOKIA	10	2	8	Reorder Stock
1-Jan-22	SONY	32	2-Jan-22	SONY	5	5	SONY	32	5	27	
1-Jan-22	LG	61	2-Jan-22	HTC	6	6	LG	61	0	61	
1-Jan-22	HTC	50	2-Jan-22	MOTOROLA	2	7	HTC	50	6	44	
1-Jan-22	MOTOROLA	30	2-Jan-22	LENOVO	15	8	MOTOROLA	30	2	28	
1-Jan-22	LENOVO	20	2-Jan-22	HONOR	20	9	LENOVO	20	15	5	Reorder Stock
	XIAOMI	25	2-Jan-22	OPPO	45	10	XIAOMI	25	0	25	
	HONOR	25	2-Jan-22			11	HONOR	25	20	5	Reorder Stock
	OPPO	50				12	OPPO	50	15	35	Reorder Stock

Proposed System

- **Modules of Inentra:**
- **Authentication**
 - Login, token generation, role validation
- **Product / Inventory Management**
 - Add, update, delete items
 - Track stock levels
 - Store supplier details, purchase history
- **Inventory Alerts**
 - Low stock alerts
 - Expiry-near alerts
 - Fast and slow-moving items detection
- **Reports & Analytics**
 - Charts and insights
 - Predictions
 - Incoming & outgoing stock summary



System Configuration

Software Requirements:

- **Frontend:** React.js
- **Backend:** Java Spring Boot
- **Database:** MySQL / MongoDB
- **Security:** JWT Authentication / Role-based Access

Hardware Requirements:

- Minimum **8GB RAM**
- Storage: 256 GB+
- Processor: Intel i3 or above
- Stable server for hosting backend



Module 1: Authentication Module

Module Description

This module ensures only authorized users can access the system.

It manages login, logout, roles, and security tokens.

Classes

- **User:** Stores user details in the DB
- **AuthController:** Handles API endpoints (/login, /register)
- **AuthService:** Authentication logic & error handling
- **JwtProvider:** Creates secure tokens
- **UserRepository:** Interacts with database for CRUD operations



The image shows a login interface for the 'Smart Inventory Management System'. At the top, a teal header bar contains the system name. Below this, a light blue box contains the login form. The form includes a prompt 'Enter your userid and password to login', a 'Userid' input field with 'admin' entered, a 'Password' input field with masked characters, and two buttons: 'Sign-in' and 'Reset'.

Database Schema (Authentication Module)

Field	Type	Description
userid	Primary Key	Unique user ID
username	Varchar	Unique username
email	Varchar	Email address
password	Varchar	Encrypted password
role	Varchar	Admin / Manager / Staff
is_active	Boolean	Account status

THANK YOU